Efficient Elicitation of Annotations for Human Evaluation of Machine Translation

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INTRODUCTION

- Workshop on Statistical Machine Translation(WMT) relies on human judgments to rank Machine Translation systems.
- Evaluators compare translations pairwise, judging which system performs better.
- However, human evaluation is expensive, subjective and noisy. Also current ranking methods have limitations
- **Goal**: Develop a method that produces more accurate rankings with less annotation effort.

#	score	range	system
1	0.638	1	UEDIN-HEAFIELD
2	0.604	2-3	UEDIN
	0.591	2-3	ONLINE-B
4	0.571	4-5	LIMSI-SOUL
	0.562	4-5	KIT
	0.541	5-6	ONLINE-A
7	0.512	7	MES-SIMPLIFIED
8	0.486	8	DCU
9	0.439	9-10	RWTH
	0.429	9-11	CMU-T2T
	0.420	10-11	CU-ZEMAN
12	0.389	12	JHU
13	0.322	13	SHEF-WPROA

THE PROBLEM & THE SOLUTION

- **Problem 1.** "Blind" Models: Current methods ignore match difficulty. A win against a weak system counts the same as a win against a strong one.
- **Problem 2.** Wasteful Sampling: Data is collected uniformly. Valuable human effort is wasted on obvious matches .
- **Problem 3.** High Cost: Generating reliable rankings requires massive amounts of data, often necessitating low-cost, non-expert annotators.
- **The Solution:** We propose adapting TrueSkil (from Xbox Live) to model relative skills and optimize data collection.

PREVIOUS MODELS

- WMT ranks MT systems using sentence-level ternary judgments of the form:

$$(i, S_1, S_2, \pi)$$

- Judgments are collected by showing evaluators:
  - 1 input sentence + reference
  - 5 randomly selected system translations
  - Annotators rank the 5 systems (ties allowed)

- Before introducing TrueSkill, two baseline approaches are used:
  - Expected Wins model
  - Hopkins & May Bayesian model

EXPECTED WINS MODEL

- Computes system scores based on how often a system wins in pairwise comparisons.

$$\text{score}(S_i) = \frac{\text{wins}(S_i) + \text{ties}(S_i)}{\text{wins}(S_i) + \text{ties}(S_i) + \text{loses}(S_i)}$$

- Ties handled inconsistently; later excluded due to bias.

$$\text{score}(S_i) = \frac{1}{|\{S\}|} \sum_{S_j \neq S_i} \frac{\text{wins}(S_i, S_j)}{\text{wins}(S_i, S_j) + \text{wins}(S_j, S_i)}$$

- Limitations
  - Treats all system comparisons as equally informative.
  - Highly sensitive to uneven sampling of system pairs.

HOPKINS & MAY MODEL

- Moves from simple counting to probabilistic modeling.
  - Assumes each system has a Relative Ability modeled as a Gaussian Distribution (μ) .

$$q_1 \sim \mathcal{N}(\mu_{S_1}, \sigma_a^2) + \mathcal{N}(0, \sigma_{obs}^2)$$
$$q_2 \sim \mathcal{N}(\mu_{S_2}, \sigma_a^2) + \mathcal{N}(0, \sigma_{obs}^2)$$

- Introduces a "Decision Radius" (d).

$$\pi = \begin{cases} < & q_1 - q_2 > d \\ > & q_2 - q_1 > d \\ = & \text{otherwise} \end{cases}$$

- If the model predicts wrong, the penalty increases linearly

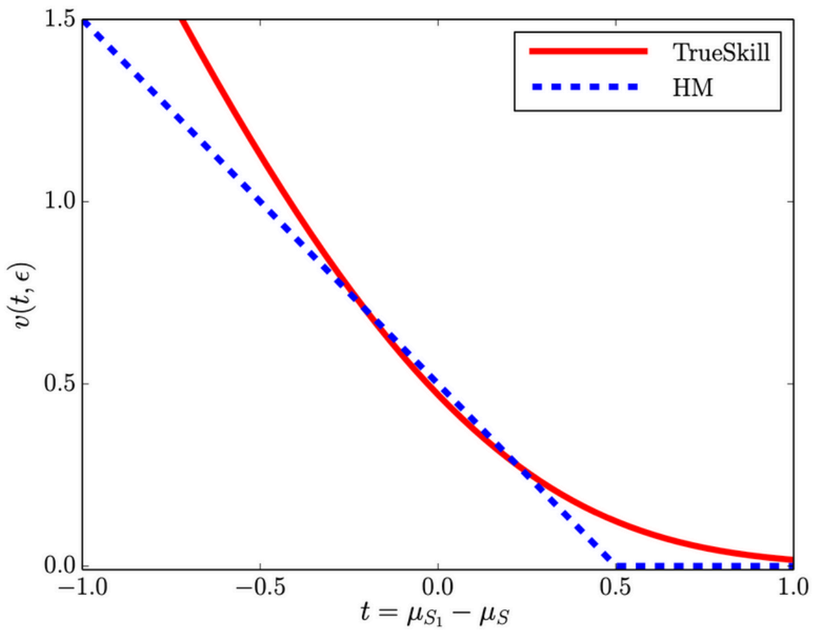
- Treats all system comparisons as equally informative.

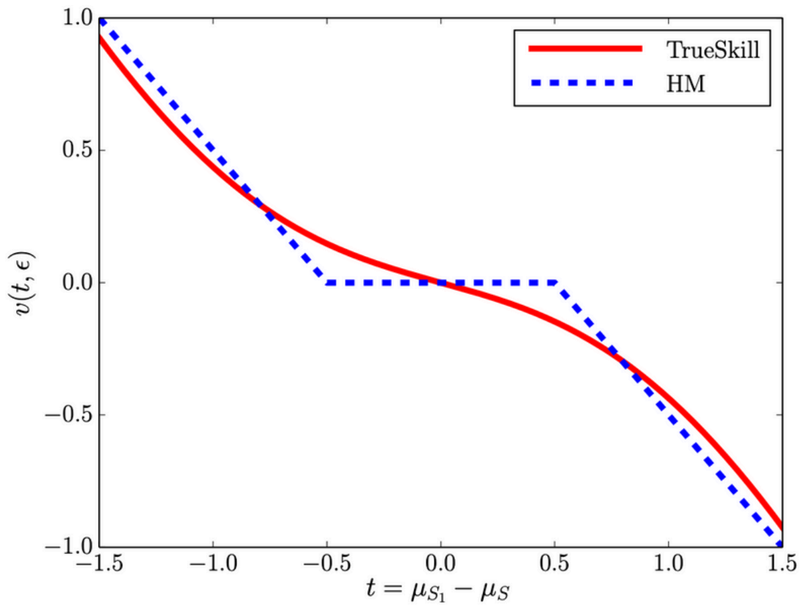
TRUESKILL

- A Bayesian ranking system originally developed for Xbox Live matchmaking.
- Instead of a single score, each system is a Gaussian Distribution.
  - μ : Average Skill.
  - σ : Uncertainty.
- The update size depends on how unexpected the outcome is
  - Surprising outcomes → large updates
  - Expected outcomes → small updates
- The ranking converges faster and handles uncertainty better than win ratios.

TRUESKILL

- Ability difference: $t = \mu_{S_1} - \mu_{S_2}$
- Surprisal-based update functions:
  - v_win,v_tie (depends on t and ε)
  - Larger when the outcome is unexpected
- Variance always decreases as more comparisons are observed





DATA SELECTION WTH TRUESKILL

- TrueSkill calculates the "Draw Probability" between any two systems.

$$\hat{p}_{\text{draw}} = \frac{1}{\exp(|\mu_a - \mu_b|)}$$

- Adaptive sampling process:

  a. Select a system with highest uncertainty
  b. Compute match probabilities with all others
  c. Sample the most informative opponent
  d. Annotate only that pair

EXPERIMENTAL SETUP

Dataset

WMT13 human evaluations

- 5-way rankings
- 10 language pairs

Training sizes: 400, 800, 1.6K, 3.2K, 6.4K comparisons

TrueSkill selects its own training data, others use uniform sampling.

Perplexity

Measure model uncertainty

Computed by integrating the Gaussian difference over outcome regions

Lower perplexity ≠ higher accuracy.

Accuracy

Measures whether the model predicts the correct outcome (<, >, =) between system pairs.

TrueSkill achieves highest accuracy, especially with small datasets.

Researcher-only data gives better results.

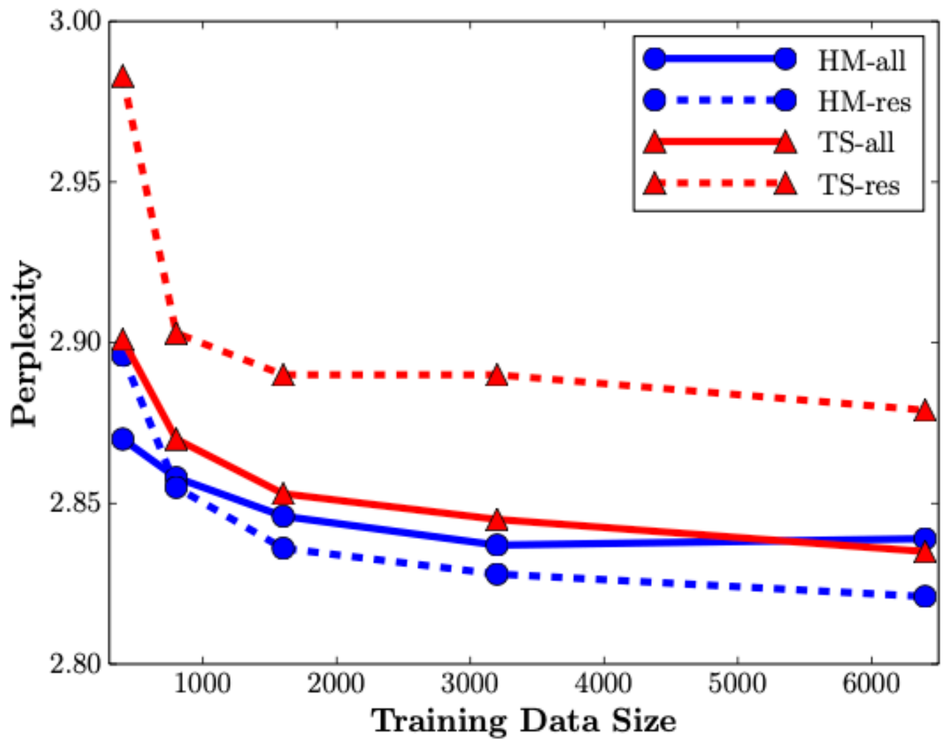
Parameter Tuning

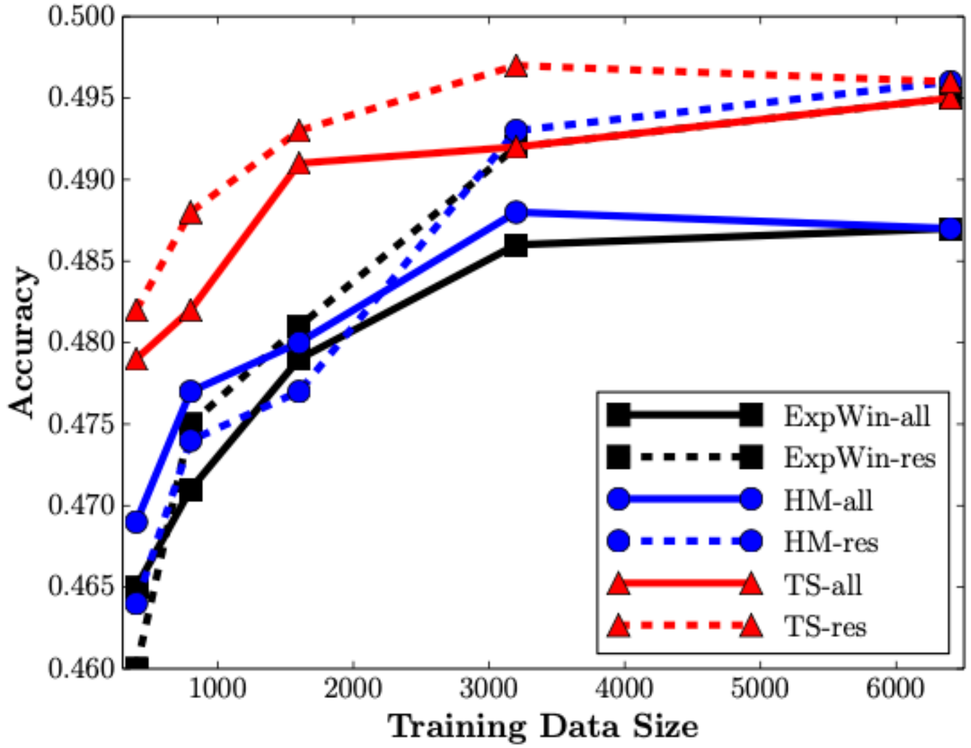
Decision radius r tuned via grid search:

- Perplexity: 0.3–0.5
- Accuracy: 0.001–0.1

RESULTS

- Perplexity decreases as training size increases, but does not align perfectly with accuracy.
- TrueSkill achieves higher accuracy, especially with smaller training sets.
- Researcher-only judgments lead to more stable and accurate models.





RESULTS

Train Size		Exp-Win	HM	TrueSkill
all	400	0.465	0.471	**0.479**
	800	0.471	0.475	**0.483**
	1600	0.479	0.477	**0.493**
	3200	0.486	0.489	**0.493**
	6400	0.487	0.490	**0.495**
res	400	0.460	0.463	**0.484**
	800	0.475	0.473	**0.488**
	1600	0.481	0.482	**0.493**
	3200	0.492	0.494	**0.497**
	6400	0.495	0.496	**0.497**
Upper Bound			0.525	

#	N=2	N=3	N=4	N=5
400	0.479	0.482	0.491	**0.492**
800	0.483	0.493	0.495	**0.495**
1600	0.493	0.492	**0.497**	0.495
3200	0.493	0.494	**0.498**	0.497
6400	0.495	**0.498**	0.498	0.498

Table 3: Accuracies when training with N-way free-for-all models, fixing the number of matches.

#	N=2	N=3	N=4	N=5
400	**0.479**	0.475	0.470	0.459
800	0.483	**0.488**	0.476	0.466
1600	**0.493**	0.488	0.481	0.481
3200	**0.493**	0.492	0.487	0.489
6400	0.495	**0.496**	0.494	0.495

Table 4: Accuracies when training with N-way free-for-all models, fixing the number of pairwise comparisons.

REDUCED DATA COLLECTION

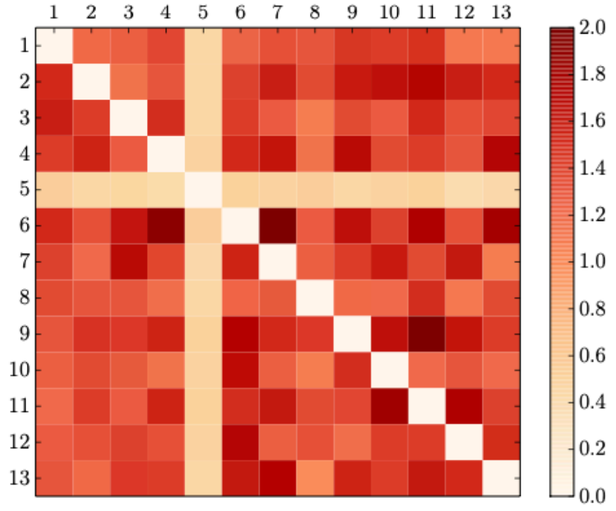


Figure 5: Heat map for the ratio of pairwise judgments across the full cross-product of systems in the WMT13 French-English translation task.

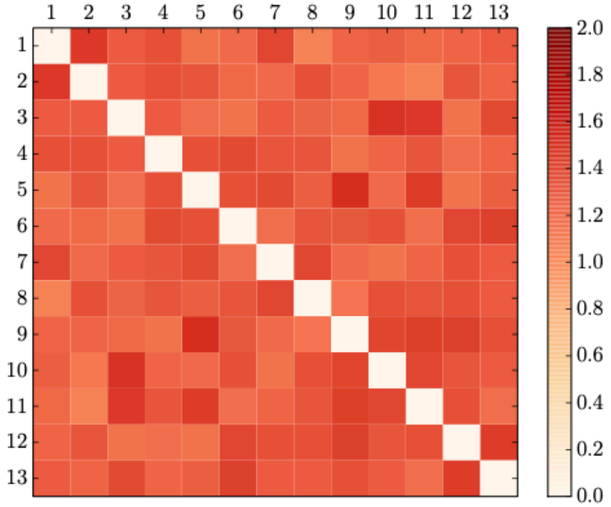


Figure 6: Heat map for the ratio of pairwise judgments across the full cross-product of systems used in the *first* 20% of TrueSkill model.

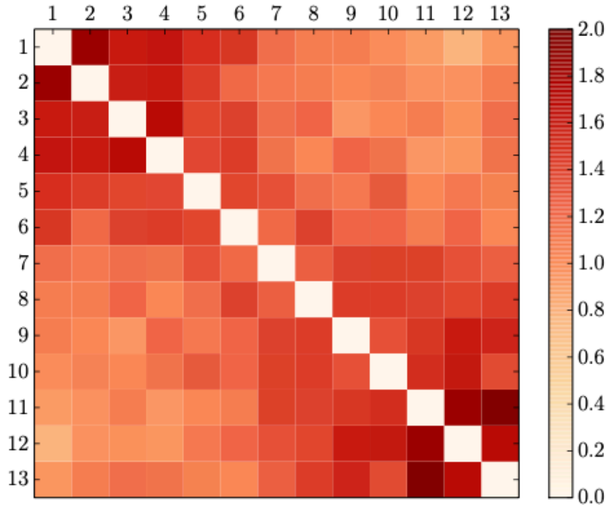
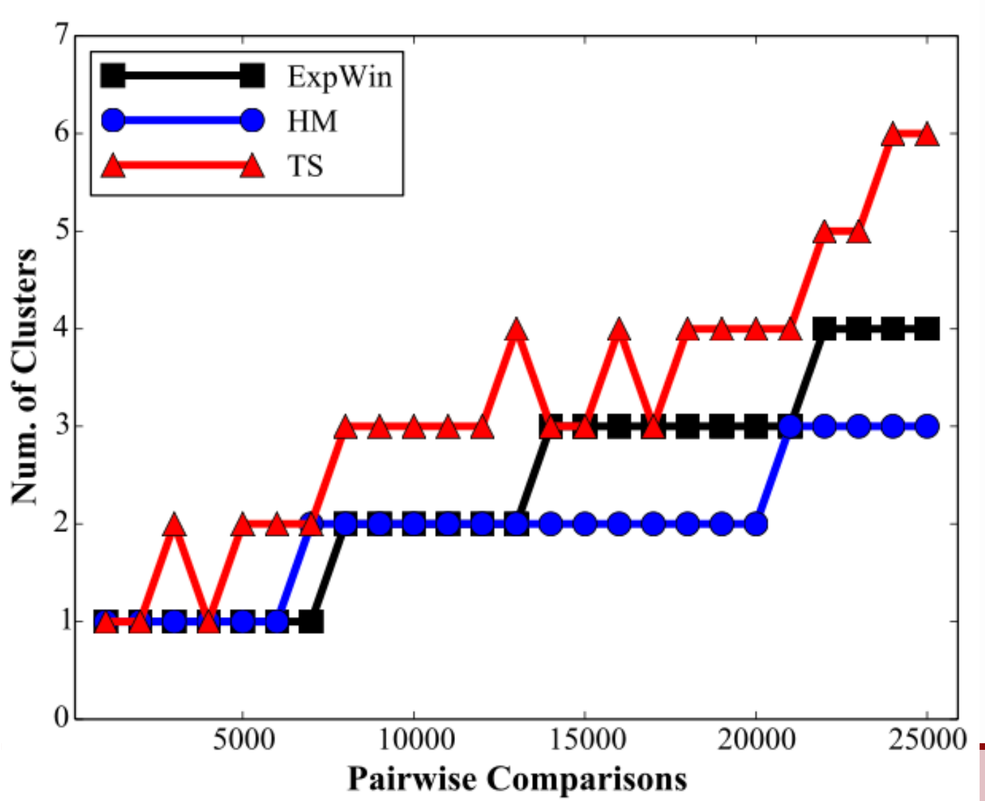
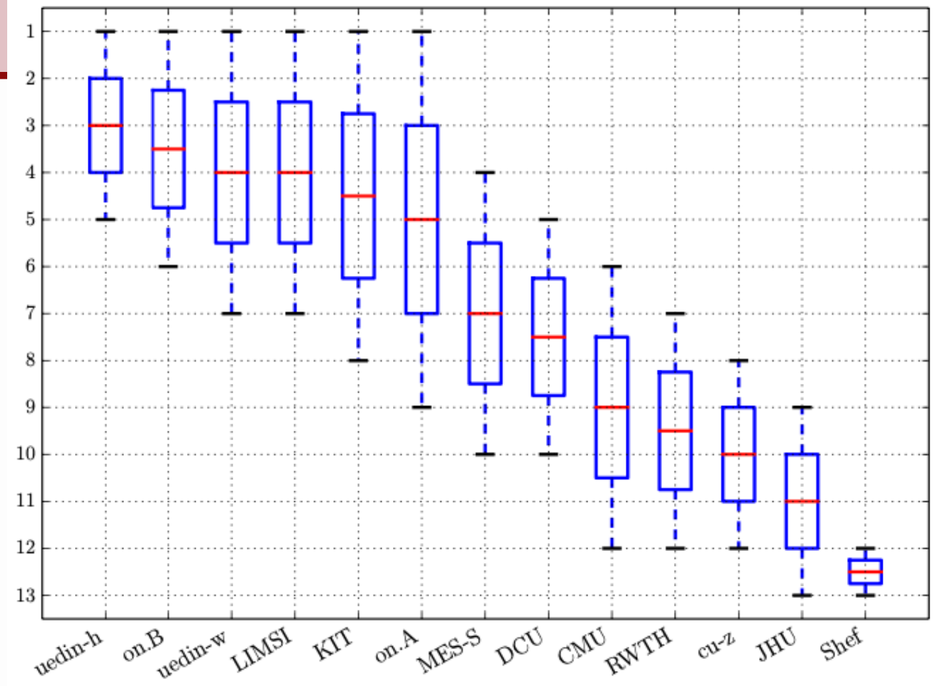


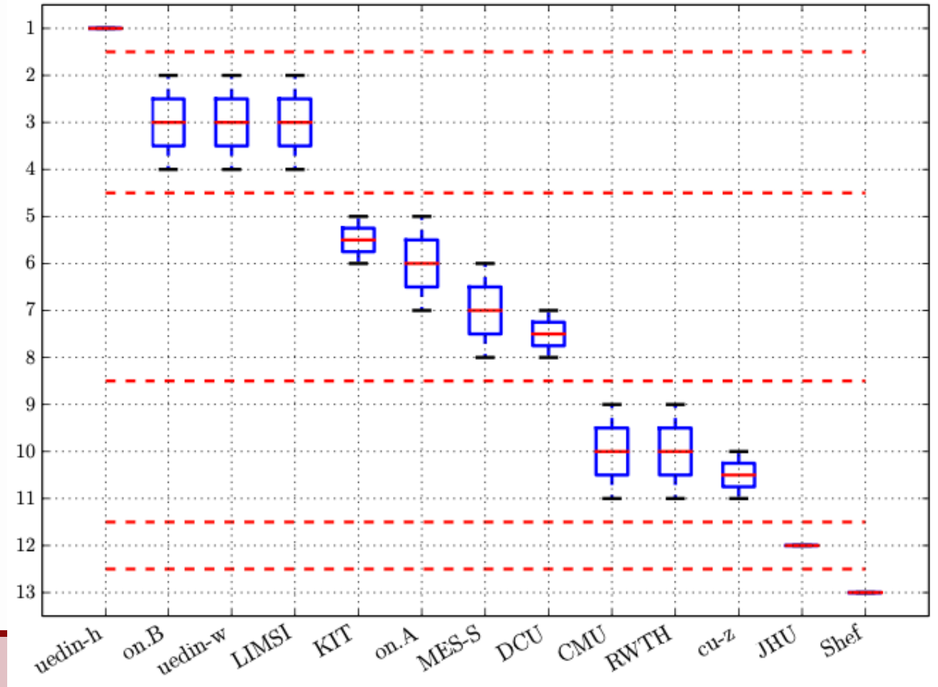
Figure 7: Heat map for the ratio of pairwise judgments across the full cross-product of systems used in the *last* 20% of TrueSkill model.

CLUSTERING

- TrueSkill produces more stable and meaningful clusters compared to other models.
- TrueSkill can form reliable clusters using far fewer annotations .







CONCLUSION

This work shows that TrueSkill is a powerful alternative to traditional WMT ranking methods. It achieves higher accuracy, models match difficulty, and enables non-uniform sampling to focus human annotation effort on the most informative system pairs. As a result, TrueSkill can significantly reduce the amount of data needed for reliable system ranking and clustering.

THANK YOU